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$$z^3 - 3z^2 - z + 3$$

$$\begin{array}{c|cccc} & 1 & 0 & 1 & -2 \\ 1 & \downarrow & 1 & 1 & 2 \\ \hline & 1 & 1 & 2 & 0 \end{array}$$

$$\Rightarrow z_1 = 1$$

$$\Rightarrow z_2 = \frac{-1 - i\sqrt{7}}{2}$$

$$\Rightarrow z_3 = \frac{-1 + i\sqrt{7}}{2}$$

En de ontbinding over $\mathbb{R}[z]$ is :

$$(z - 1)(z^2 + z + 2)$$

References